School of Electrical Engineering & Telecommunications

ELEC1111 Workshop 5

Q1 (Sample exam question)

The circuit in Figure 1 is operating under DC steady state conditions.

i) If R = 10 Ω, determine the voltage across the capacitor, v_c, and the current through the inductor, i_L.

ii) Calculate the energy stored in the capacitor and the inductor in this situation (i.e., when R = 10 Ω).

iii) Calculate the value of R that will make the energy stored in the capacitor the same as the energy stored in the inductor.

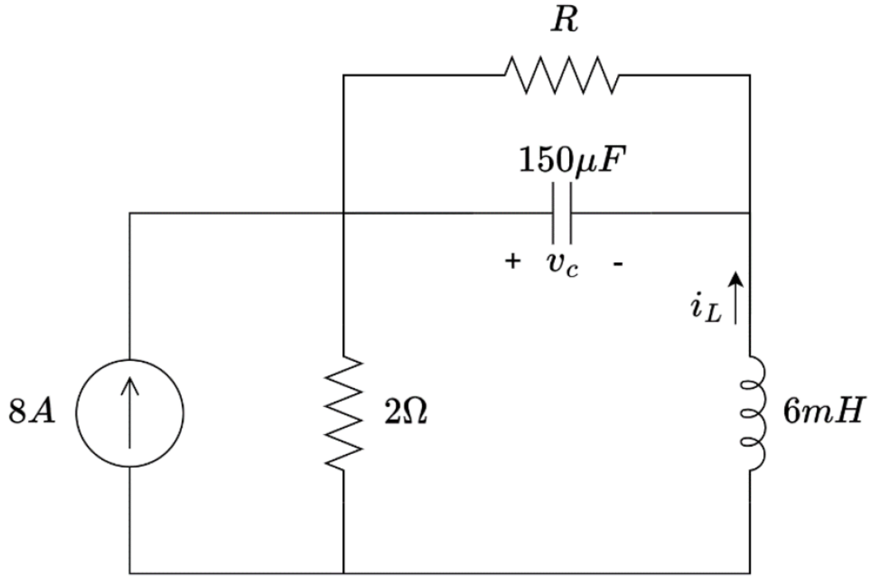

Figure 1

Answers: i) v_c = 13.33 V, i_L = -1.333 A, ii) W_c = 0.0133 J, W_L = 0.00533 J, iii) R = 6.32 Ω

Q2 (2021 T1 Final Exam Q2b)

Switch 1 (SW1) in Figure 2 was in position a for a long time before moving to position b at t=0. After 0.1 seconds, the voltage source is disconnected from the circuit by opening Switch 2 (SW2).

i) Calculate $i_0(t)$ for $0 \le t < 0.1$s.

ii) Calculate $i_0(t)$ for $t \ge 0.1$s.

iii) Calculate the value of i_0 at $t = 0.2$s.

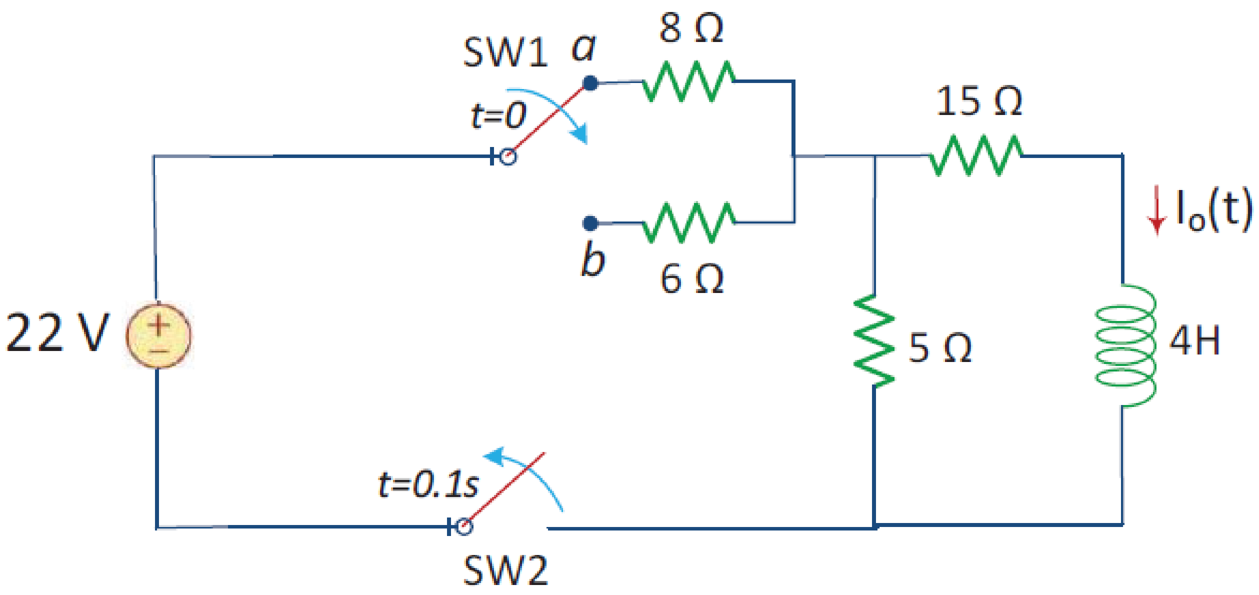

Figure 2

Answers: i) $i_0(t) = 0.564 - 0.096e^{-4.43t}\ A$, ii)$i_0(t) = 0.5025e^{-5(t-0.1)}\ A$, iii) $i_0(0.2) = 0.305\ A$

Q3 (2020 T1 Final Exam Q2a)

A magnetically controlled switch is called a relay. A relay is essentially an electromagnetic device used to open or close a switch that controls another circuit (see Figure 3(a)). The coil circuit is an RL circuit like that in Figure 3(b), where R and L are the resistance and inductance of the coil.

A certain relay has a resistance of 200Ω and an inductance of 500 mH. The relay contacts close when the current through the coil reaches 175 mA. What time elapses between the application of 110V to the coil (when S1 closes at t = 0) and contact closure?

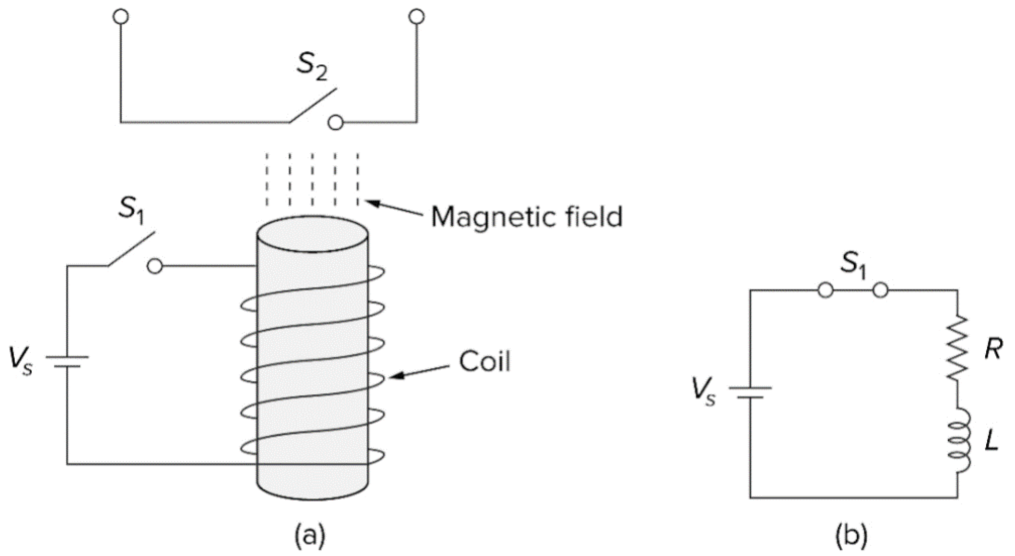

Figure 4

Answer: 957.5 µs

Q4 (Sample exam question)

The switch in the RL circuit shown in Figure 4 was closed for a long time before opening at t = 0 s.

i) Derive the expression for $i_L(t)$ for t ≥ 0 s.

ii) Calculate the value of $v_L(t)$ at t = 5 ms.

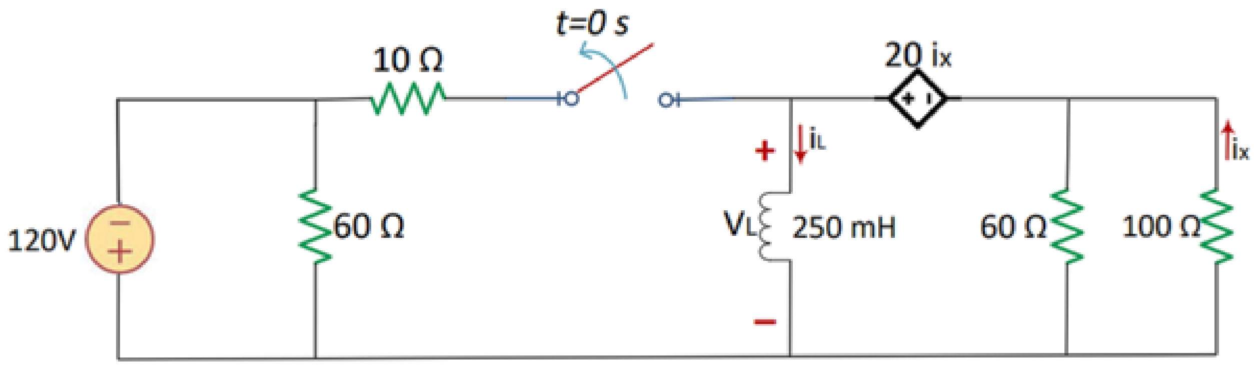

Figure 4

Answers: i) $i_L(t) = -12e^{-120t}\ A$, ii) $v_L(t = 0.005) = 197.57\ V$